

End-to-End STEM Subject Classification

Leakage-safe ML workflow | Python | SQL | scikit-learn | CI/CD

Conditional Go - Validated Offline Prototype

0.8755

FINAL MACRO-F1

95% CI 0.8471-0.8996

0.8778

FINAL ACCURACY

553/630 correct

0.8807

GROUPED-CV MACRO-F1

+/- 0.0171 before test access

PIN DATA

VALIDATE

GROUP SPLIT

TUNE

LOCK TEST

DIAGNOSE

Problem and Data

Can question text alone route English STEM benchmark questions into mathematics, physics, chemistry, or biology? The pipeline pinned 4499 records, removed 91 exact duplicates, and retained 4408 controlled rows.

Leakage Controls

- 0 duplicate groups cross development and final test.
- TF-IDF is fitted inside every training fold.
- 72 parameter settings use grouped CV only.
- Selection hashes are persisted before one-time test evaluation.

Engineering Evidence

Importable Python package and CLI; 10 named SQL analyses; artifact hashes; automated tests; clean-room install; GitHub Actions; numerical gates; and a visually audited 15-page report.

Findings

- Math is strongest at F1 0.9361.
- Chemistry is weakest at F1 0.8391.
- Questions below 100 characters have 19.7% error.
- The learning curve still improves with more development data.

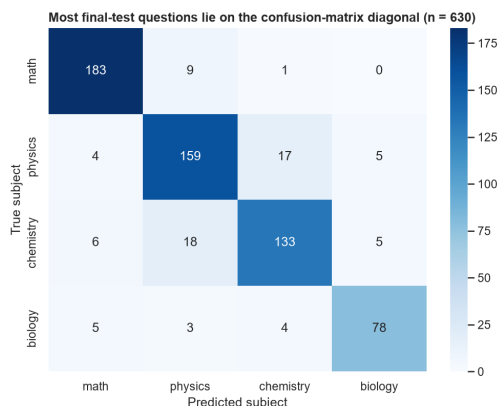
Capability Boundaries

- Benchmark evidence, not student-behavior or manufacturing validation.
- LinearSVC margins are not calibrated probabilities.
- Source-subject confounding leaves cross-source transfer unverified.
- Production readiness and automated decisions are not claimed.

Decision

Proceed as an offline four-subject routing prototype. Next valid evidence requires new multi-source data or a separately frozen manufacturing case, not reuse of the current final test.

Final-Test Behavior



Locked model only | n = 630 | verified metric artifact

Five-Minute Review Path

- 1 Read this one-page decision summary.
- 2 Open selection_decision.json for the lock evidence.
- 3 Inspect confusion and error-slice artifacts.
- 4 Review limits and the not-feasible holdout.
- 5 Run stem-analytics run-all to show reproducibility.

Core portfolio signal

The result matters, but the stronger evidence is disciplined data control, honest evaluation, reproducible engineering, and decisions tied to failure analysis.